

LINUX COMMANDS

Module 2 • Category-Wise Practical Guide

Linux, Networking & Cloud Infrastructure

A practical command reference organized around the Module 2 requirements: Linux CLI, files and permissions, users and groups, process management, services, and SSH keys.

Goal: Understand what each command does, when to use it, and how it connects to the Azure Linux VM practical.

Md Shahil

1. Module 2 Linux Command Map

The module specifically requires Linux CLI operations, files/permissions, users/groups, process management, services, and SSH keys.

Category	Purpose	Core commands
CLI & navigation	Move around and inspect the Linux system	pwd, ls, cd, clear, history, man
Files & directories	Create, view, copy, move and remove data	touch, mkdir, cp, mv, rm, cat, less, head, tail
Permissions & ownership	Control read, write and execute access	ls -l, chmod, chown, chgrp, umask
Users & groups	Manage accounts and memberships	whoami, id, useradd, passwd, usermod, groupadd
Processes	Inspect and control running processes	ps, top, pgrep, kill, pkill
Services	Manage background services	systemctl status/start/stop/restart/enable
SSH & keys	Secure remote access to Linux systems	ssh, ssh-keygen, ssh-copy-id, ~/.ssh

Practical chain: SSH → Linux configuration → permissions/users → service management → Nginx → connectivity test.

2. Linux CLI & Navigation

These commands help you understand where you are, inspect directories, move between locations, and get help.

Command	Purpose	Example
pwd	Print current working directory.	pwd
ls	List files and directories.	ls
ls -l	Detailed listing including permissions and ownership.	ls -l
ls -la	Include hidden files.	ls -la
cd	Change directory.	cd /var/www/html
cd ..	Move to parent directory.	cd ..
cd ~	Go to the current user's home directory.	cd ~
clear	Clear the terminal.	clear
history	Show previous commands.	history
man	Open a command manual.	man ls
which	Show executable path when available.	which nginx

3. Files & Directories

Day-to-day file operations used when configuring a Linux server and deploying applications.

Command	Purpose	Example
mkdir	Create a directory.	mkdir app
mkdir -p	Create parent directories as needed.	mkdir -p app/config/logs
touch	Create an empty file.	touch notes.txt
cat	Display file contents.	cat /etc/hostname
less	View a file page by page.	less /var/log/syslog
head	Show the beginning of a file.	head -n 20 file.txt
tail	Show the end of a file.	tail -n 20 file.txt
tail -f	Follow a growing log.	tail -f /var/log/nginx/access.log

Command	Purpose	Example
cp	Copy files/directories.	cp index.html backup.html
cp -r	Copy a directory recursively.	cp -r app app_backup
mv	Move or rename.	mv old.conf new.conf
rm	Remove a file.	rm notes.txt
rm -r	Remove a directory recursively.	rm -r old_app
find	Search for files/directories.	find /var/www -name '*.html'

4. File Permissions & Ownership

Linux permissions control access for the owner, group, and others.

Command	Purpose	Example
ls -l	Inspect permissions and ownership.	ls -l file.txt
chmod	Change permissions.	chmod 644 index.html
chmod +x	Add execute permission.	chmod +x script.sh
chmod -x	Remove execute permission.	chmod -x script.sh
chown	Change owner.	sudo chown ubuntu file.txt
chown user:group	Change owner and group.	sudo chown ubuntu:www-data file.txt
chgrp	Change group ownership.	sudo chgrp www-data file.txt
umask	Show/set default permission mask.	umask

Permission model: owner/user + group + others; r = read, w = write, x = execute.

Common modes: 644 = regular files; 755 = directories/executable scripts; 600 = private files.

5. Users & Groups

User and group management provides controlled access to a Linux server.

Command	Purpose	Example
whoami	Show current username.	whoami
id	Show UID, GID and memberships.	id
who	Show logged-in users.	who
useradd	Create a user.	sudo useradd -m devuser
passwd	Set/change a user's password.	sudo passwd devuser
usermod -aG	Add a user to a supplementary group.	sudo usermod -aG sudo devuser
groupadd	Create a group.	sudo groupadd developers
groups	Show groups for a user.	groups devuser
userdel	Delete a user account.	sudo userdel devuser

6. Process Management

Processes are running programs. These commands help inspect activity and stop processes when necessary.

Command	Purpose	Example
ps	Show processes in the current context.	ps
ps aux	Detailed list of running processes.	ps aux

Command	Purpose	Example
top	Interactive CPU/memory/process view.	top
pgrep	Find process IDs by name/pattern.	pgrep nginx
kill	Send a signal to a process ID.	sudo kill 1234
pkill	Signal processes matching a name/pattern.	sudo pkill nginx

7. Linux Services & systemd

systemd commonly manages services on modern Linux distributions such as Ubuntu.

Command	Purpose	Example
systemctl status	Check service state.	sudo systemctl status nginx
systemctl start	Start a service now.	sudo systemctl start nginx
systemctl stop	Stop a service now.	sudo systemctl stop nginx
systemctl restart	Restart a service.	sudo systemctl restart nginx
systemctl enable	Start automatically at boot.	sudo systemctl enable nginx
systemctl disable	Prevent automatic boot start.	sudo systemctl disable nginx
systemctl is-active	Check whether a service is active.	systemctl is-active nginx
systemctl is-enabled	Check boot-enabled state.	systemctl is-enabled nginx

8. SSH & SSH Keys

SSH provides secure remote access to a Linux server and is central to the Azure VM practical.

Command	Purpose	Example
ssh	Connect to a remote SSH server.	ssh azureuser@203.0.113.10
ssh -i	Use a specific private key.	ssh -i ~/.ssh/azure_vm_key azureuser@203.0.113.10
ssh-keygen	Generate an SSH key pair.	ssh-keygen -t ed25519
ssh-keygen -lf	Display a public-key fingerprint.	ssh-keygen -lf ~/.ssh/id_ed25519.pub
ssh -v	Verbose SSH troubleshooting output.	ssh -v azureuser@203.0.113.10
ssh-copy-id	Copy a public key to a remote account when supported.	ssh-copy-id user@server

Security rule: keep the private key on your local machine; never publish it to GitHub or share it.

```
ssh -i ~/.ssh/azure_vm_key azureuser@<VM_PUBLIC_IP>
```

9. Module 2 Azure Linux VM Practical

Use this sequence to build the evidence for the module's practical requirement: Create VM → Configure Linux → Secure Access → Deploy Service → Test Connectivity.

Step	Action	Evidence
1	Create/use an Azure Linux VM.	Azure VM overview/configuration screenshot
2	Verify network access rules.	VNet/subnet/NSG/port screenshot
3	Connect using SSH.	Successful SSH terminal screenshot
4	Run Linux checks.	whoami, hostname, pwd, ip addr
5	Install Nginx.	Installation command/output screenshot
6	Manage the service.	systemctl status nginx

Step	Action	Evidence
7	Test connectivity.	Browser Nginx page or curl response

Example Nginx deployment

```
sudo apt update sudo apt install nginx -y sudo systemctl enable --now nginx sudo systemctl status nginx
```

Local service test:

```
curl http://localhost
```

For external access, Azure networking must allow the required traffic to reach the service. Keep exposed ports to the minimum required.

10. Quick Troubleshooting

Question/problem	Useful checks
Where am I?	pwd
What is my username?	whoami / id
What files are here?	ls -la
What permissions does this file have?	ls -l filename
Who owns the file?	ls -l filename / stat filename
Is Nginx running?	systemctl status nginx
What process is using Nginx?	ps aux / pgrep nginx
What IP addresses does the VM have?	ip addr
Can I reach the local web service?	curl http://localhost
Why is SSH failing?	ssh -v user@host; check VM, NSG and SSH service

11. Practice Checklist

- Navigate the filesystem using pwd, ls and cd.
- Create, copy, move, inspect and remove files/directories.
- Explain read/write/execute permissions and use chmod safely.
- Use chown and understand ownership.
- Explain users vs groups and check membership with id/groups.
- Inspect running processes and understand process IDs.
- Start, stop, restart and check a service with systemctl.
- Generate and use an SSH key pair.
- Connect to an Azure Linux VM using SSH.
- Install and verify Nginx.
- Test the service locally and understand the network path for external access.

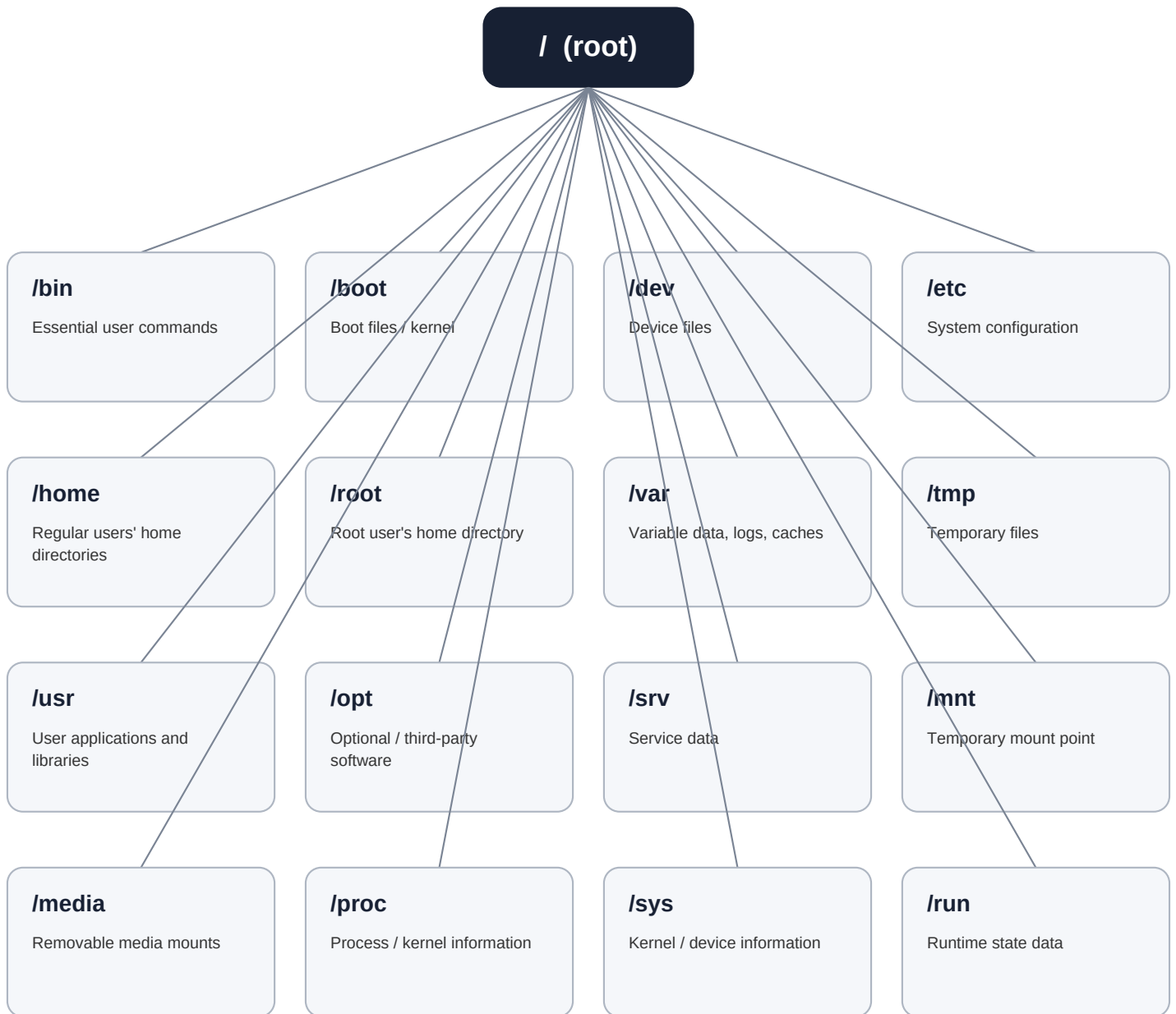
12. Suggested GitHub Evidence Structure

```
Module-2-Linux-Networking-Azure/ ■■■ README.md ■■■ linux/ ■ ■■■ linux-notes.md ■■■ networking/ ■
■■■ networking-notes.md ■■■ diagrams/ ■ ■■■ linux-directory-structure.png ■■■ screenshots/ ■■■
azure-linux-vm.png ■■■ ssh-success.png ■■■ nginx-installation.png ■■■ nginx-running.png
```

This PDF focuses on the Linux command portion of Module 2. Keep IP addressing, DNS, ports, firewall and Azure NSG as a separate networking notes section/file.

Linux File System Architecture

Reference diagram for understanding common Linux directories and their typical purpose.



Important:

/etc is commonly used for configuration; /var contains changing data such as logs; /home contains user data.

/proc, /sys and /run are virtual/runtime areas provided by the Linux system rather than ordinary storage directories.